Latent Denoising Diffusion Probabilistic Models: Implementation, Acceleration, and Conditional Generation on ImageNet-100

Rakshanda Abhinandan **Pranav Bhosale** **Swetleena Guha**

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Group Number: 11
Carnegie Mellon University

Abstract

Diffusion models have emerged as one of the most powerful classes of generative models, surpassing GANs in stability and image fidelity. In this project, we implement a complete latent diffusion pipeline from scratch, progressing from a pixel-space Denoising Diffusion Probabilistic Model (DDPM) to an accelerated Denoising Diffusion Implicit Model (DDIM) sampler, then to a Latent Diffusion Model (LDM) operating in a pre-trained VAE latent space, and finally to class-conditional generation via Classifier-Free Guidance (CFG) on the ImageNet-100 dataset (100 classes, 128×128 resolution). We further explore advanced training techniques including a cosine noise schedule and Exponential Moving Average (EMA) of model weights. We evaluate each configuration using Fréchet Inception Distance (FID) and Inception Score (IS) on 5,000 generated samples. Our final Latent DDPM with CFG and advanced training yields visibly coherent class-conditional samples and achieves the best FID/IS among all configurations tested, validating the design choices motivated by the diffusion literature.

1 Introduction

1.1 Problem Statement and Motivation

Generating realistic images from random noise is a fundamental problem in machine learning and computer vision. Generative models that can learn the underlying distribution of natural images and synthesize new, photorealistic samples are central to applications including content creation, data augmentation, simulation, and scientific imaging. Traditional generative approaches such as Generative Adversarial Networks (GANs) often suffer from training instability and mode collapse. Diffusion models offer a more stable alternative by modeling generation as a gradual noise corruption and learned denoising process (Ho et al., 2020).

The objective of this project is to design, implement, and evaluate a class-conditional latent diffusion model capable of generating 128×128 images on the ImageNet-100 dataset (100 classes, $\sim$130,000 images). The implementation is built from scratch and incrementally extended through four stages: (i) a baseline pixel-space DDPM, (ii) a DDIM sampler for inference acceleration, (iii) a latent-space DDPM operating on top of a pre-trained Variational Autoencoder (VAE), and (iv) class-conditional generation using Classifier-Free Guidance (CFG). We further explore advanced training techniques to improve sample quality.

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1.2 Quantitative and Qualitative Objectives

Quantitative: Minimize the mean squared error (MSE) between predicted and true noise during training, and minimize Fréchet Inception Distance (FID) while maximizing Inception Score (IS) on 5,000 generated samples. **Qualitative:** Generate visually meaningful, class-coherent images that resemble the ImageNet-100 data distribution.

1.3 Contributions

This work delivers (1) a from-scratch implementation of DDPM, DDIM, Latent DDPM, and CFG; (2) systematic evaluation of each component using FID and IS on a fixed validation reference; (3) an ablation over CFG guidance scales; and (4) exploration of advanced training techniques (cosine noise schedule, EMA weights) that improve generation quality.

2 Related Work and Background

2.1 Literature Review

We organize prior work into three lines that directly inform our design.

Foundational diffusion models. Ho et al. (2020) introduced DDPM, formulating image generation as a fixed-variance Markov chain that adds Gaussian noise in the forward process and a learned reverse process trained via a simplified denoising objective. This established diffusion models as a stable alternative to GANs and provided the architectural recipe (U-Net with sinusoidal time embeddings) that we build on.

Sampling acceleration. Song et al. (2021) proposed DDIM, a non-Markovian generalization of DDPM that admits a deterministic update rule and produces high-quality samples in 50–100 steps rather than 1,000. DDIM motivates our acceleration step in Section 4.

Latent and conditional diffusion. Rombach et al. (2022) introduced Latent Diffusion Models, which run the diffusion process in a compressed VAE latent space, dramatically reducing compute. Ho & Salimans (2022) introduced Classifier-Free Guidance, which enables conditional generation by jointly training a single model on conditional and unconditional objectives, avoiding the need for an external classifier as in Dhariwal & Nichol (2021). Together these works form the basis for our Sections 5 and 6.

2.2 Background

The choice of FID (Heusel et al., 2017) and IS (Salimans et al., 2016) as evaluation metrics, and the choice of a U-Net backbone with attention, are directly motivated by these foundational papers, which use the same metrics and architecture family. Our baseline (pixel DDPM) and each extension (DDIM, LDM, CFG) maps one-to-one to a cited reference, ensuring the design space is grounded in the literature.

3 Methodology

3.1 Model Description

U-Net backbone. Our denoising network is a U-Net (Ronneberger et al., 2015) that takes a noisy input x_t (or latent z_t) and a timestep t, and predicts the noise $\epsilon_\theta(x_t, t)$. The encoder downsamples the input through residual blocks with channel multipliers $[1, 2, 2, 4]$, the bottleneck applies self-attention at the lowest resolution, and the decoder mirrors the encoder with skip connections. Time is injected as a sinusoidal positional embedding passed through a 2-layer MLP and added to each residual block. For the conditional model, a learned class embedding (with random dropout for CFG) is added to the time embedding.

Output map shapes and parameter counts. For pixel-space DDPM the input/output map is $3 \times 128 \times 128$. For Latent DDPM the VAE compresses to a $4 \times 16 \times 16$ latent ($8\times$ spatial reduction), so the U-Net operates on $4 \times 16 \times 16$ tensors. The pixel-space U-Net has $\sim$36M parameters; the

latent U-Net has $\sim$28M parameters; the pre-trained VAE adds $\sim$84M (frozen) parameters. The class embedder uses a 100-token table of 512-d embeddings ($\sim$51K params).

Pipeline. For pixel DDPM: $x_t \to \text{U-Net}(x_t, t) \to \hat{\epsilon} \to \text{scheduler} \to x_{t-1}$. For Latent DDPM with CFG: $x_0 \xrightarrow{\text{VAE enc}} z_0 \to z_t \to \text{U-Net}(z_t, t, y) \to \hat{\epsilon}_{\text{cfg}} \to z_{t-1} \to \cdots \to z_0 \xrightarrow{\text{VAE dec}} \hat{x}_0$.

3.2 Dataset

We use **ImageNet-100**, a 100-class subset of ImageNet with approximately 130,000 training images and a held-out validation set, all resized to 128×128. Images are normalized to $[-1, 1]$ to match the diffusion noise schedule's assumed support. Training uses a batch size of 32 (pixel space) or 128 (latent space, due to the smaller spatial dimension). For each training step we (i) sample a minibatch of (x_0, y), (ii) sample a uniform timestep $t \sim \mathcal{U}\{1, \ldots, T\}$ per example, (iii) sample $\epsilon \sim \mathcal{N}(0, I)$, (iv) form $x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon$, and (v) update parameters by gradient descent on the loss in Equation (3). Dataset reference: Deng et al. (2009).

3.3 DDPM: Forward and Reverse Processes

The forward process is a fixed Markov chain that adds Gaussian noise according to a variance schedule $\beta_1, \ldots, \beta_T$:

$$q(x_t \mid x_{t-1}) = \mathcal{N}\Big(x_t; \sqrt{1 - \beta_t}\, x_{t-1},\, \beta_t I\Big)\,, \tag{1}$$

which admits a closed-form marginal $q(x_t \mid x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t} x_0, (1 - \bar{\alpha}_t) I)$ where $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$. The reverse process $p_\theta(x_{t-1} \mid x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t))$ is parameterized so the network predicts noise:

$$\mu_\theta(x_t, t) = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}}\, \epsilon_\theta(x_t, t) \right). \tag{2}$$

3.4 Loss Function

Following Ho et al. (2020), we use the simplified denoising objective, which is the MSE between the true noise ϵ and the network prediction ϵ_θ:

$$\mathcal{L}_{\text{simple}}(\theta) = \mathbb{E}_{t, x_0, \epsilon} \Big[\big\| \epsilon - \epsilon_\theta\big(\sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1 - \bar{\alpha}_t}\, \epsilon,\, t\big) \big\|^2 \Big]\,. \tag{3}$$

We optimize with AdamW (lr $= 2 \times 10^{-4}$, weight decay $= 0.01$, $\beta_1 = 0.9$, $\beta_2 = 0.999$), a linear warmup of 1,000 steps, and gradient clipping at norm 1.0. For pixel-space DDPM we use $T = 1000$ timesteps with a linear β-schedule from 10^{-4} to 0.02.

3.5 Evaluation Metrics

Fréchet Inception Distance (FID). FID compares Inception-v3 pool3 features ($d = 2048$) of generated and reference images, modeling each as a multivariate Gaussian:

$$\text{FID} = \|\mu_r - \mu_g\|^2 + \text{Tr}\Big(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2}\Big)\,. \tag{4}$$

Lower FID indicates better fidelity and diversity. Here μ_r, Σ_r are computed on the ImageNet-100 validation set and μ_g, Σ_g on 5,000 generated images.

Inception Score (IS). IS measures both quality (low conditional entropy $p(y \mid x)$) and diversity (high marginal entropy $p(y)$):

$$\text{IS} = \exp(\mathbb{E}_x[D_{\text{KL}}(p(y \mid x) \,\|\, p(y))])\,. \tag{5}$$

Higher IS is better. Both metrics are computed using the `torchmetrics` library to ensure standardized, reproducible evaluation across all configurations.

3.6 Experimental Depth and Iteration

We trained and evaluated five distinct configurations (DDPM, DDPM+DDIM, Latent DDPM, Latent DDPM+CFG, Latent DDPM+CFG with advanced techniques), and ran additional ablations over (a) DDIM step counts $\{1000, 100, 50, 25\}$, (b) CFG guidance scales $w \in \{0, 1, 2, 3, 5, 7\}$, and (c) noise schedules (linear vs. cosine). This represents over a dozen training/inference runs in total. Each model checkpoint was saved every 10 epochs, and we report the best-FID checkpoint per configuration. Training logs (loss curves, EMA decay, learning rate schedule) were tracked with TensorBoard for every run.

4 DDIM Scheduler for Inference Acceleration

A central inefficiency of DDPM is that inference requires traversing the full $T = 1000$-step Markov chain, which is prohibitively slow for evaluating large batches of generated samples. Song et al. (2021) resolve this by deriving a non-Markovian family of forward processes that share the same training-time marginals as DDPM, but admit a deterministic reverse update with arbitrarily fewer steps. The DDIM update rule is:

$$x_{t-1} = \sqrt{\bar{\alpha}_{t-1}} \underbrace{\left(\frac{x_t - \sqrt{1-\bar{\alpha}_t}\, \epsilon_\theta(x_t, t)}{\sqrt{\bar{\alpha}_t}} \right)}_{\hat{x}_0\text{-prediction}} + \sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2}\, \epsilon_\theta(x_t, t) + \sigma_t\, z_t, \tag{6}$$

where $z_t \sim \mathcal{N}(0, I)$ and σ_t controls stochasticity. Setting $\sigma_t = 0$ yields a fully deterministic sampler. We implement DDIM by (i) selecting a uniformly-spaced sub-sequence of timesteps $\tau_1 < \cdots < \tau_S$ from $\{1, \ldots, T\}$, (ii) re-using the DDPM-trained ϵ_θ without retraining, and (iii) applying Equation (6) at each $\tau_i \to \tau_{i-1}$. The same trained network can therefore be sampled at multiple step counts at no additional training cost.

5 Latent Diffusion Model

Pixel-space diffusion at 128×128 is computationally expensive: each U-Net forward pass touches $3 \cdot 128^2 = 49{,}152$ inputs. Rombach et al. (2022) observed that natural images contain significant perceptually-irrelevant detail that wastes diffusion capacity, and proposed running diffusion in a compressed latent space produced by a frozen VAE. We use a pre-trained VAE that maps $3 \times 128 \times 128 \mapsto 4 \times 16 \times 16$, an $64\times$ reduction in tokens. The VAE encoder produces a latent $z_0 = \mathcal{E}(x_0)$, which is scaled by a fixed factor of 0.1845 to yield approximately unit variance (matching the noise schedule's assumption); the decoder reconstructs $\hat{x}_0 = \mathcal{D}(z_0/0.1845)$. Diffusion training and sampling then proceed exactly as in Section 3.3, but with z_t replacing x_t. We do *not* fine-tune the VAE.

6 Classifier-Free Guidance

Unconditional generation produces diverse but uncontrolled samples. To generate images of a specified ImageNet-100 class, we adopt Classifier-Free Guidance (Ho & Salimans, 2022), which trains a single network jointly on conditional and unconditional objectives. During training, the class label y is replaced with a learned null token $\varnothing$ with probability $p_{\text{drop}} = 0.1$. At inference, each denoising step performs two forward passes and combines them:

$$\hat{\epsilon}_\theta(x_t, t, y) = \epsilon_\theta(x_t, t, \varnothing) + w \cdot \big(\epsilon_\theta(x_t, t, y) - \epsilon_\theta(x_t, t, \varnothing)\big), \tag{7}$$

where w is the guidance scale. $w = 0$ recovers unconditional sampling, $w = 1$ recovers standard conditional sampling, and $w > 1$ amplifies the class signal at the cost of diversity. We sweep $w \in \{0, 1, 2, 3, 5, 7\}$ in Section 9.

7 Baseline Selection, Reproduction, and Extensions

7.1 Baseline Selection and Justification

Our baseline is a pixel-space DDPM, which directly follows the original DDPM formulation of Ho et al. (2020). We chose this baseline because (i) it is the canonical diffusion model and is the reference

against which DDIM, LDM, and CFG were originally proposed; (ii) it isolates the effect of each subsequent extension (acceleration, latent compression, conditioning) without confounds; and (iii) all reported improvements in the literature are measured against an equivalent pixel DDPM. The same evaluation pipeline (5,000-sample FID/IS against ImageNet-100 validation) is used for every configuration to ensure systematic, comparable evaluation.

7.2 Baseline Reproduction Evidence

We verified our baseline DDPM implementation by reproducing the expected qualitative trajectory of pixel-space diffusion training: (i) the simple loss $\mathcal{L}_{\text{simple}}$ decreases monotonically over training (see Figure **??**, training loss panel), (ii) the model produces noise at epoch 1 and progressively recovers low-frequency structure (color blobs, rough silhouettes) by epoch 100, and (iii) at $T = 1000$ DDPM sampling steps the unconditional samples have FID/IS within the range expected for a U-Net of this size trained for the budget used. The training-loss decrease and qualitative progression match the behavior described in Ho et al. (2020) and the writeup's note that "you can expect legible results after a few hundred epochs". As a further sanity check, we verified that DDIM with $\sigma_t = 0$ and $S = T$ steps produces samples nearly identical to DDPM's (deterministic-DDIM at full steps is theoretically equivalent to running the same trained model with the DDPM update).

7.3 Implemented Extensions

We implemented four extensions on top of the DDPM baseline: (1) DDIM sampler (Section 4); (2) Latent DDPM with frozen VAE (Section 5); (3) Classifier-Free Guidance (Section 6); and (4) advanced training techniques (Section 8). Each extension is evaluated independently and in combination so that the contribution of each can be attributed.

8 Advanced Training Techniques

Beyond the four required components, we explored two advanced training techniques.

Cosine noise schedule. The default linear β-schedule destroys signal too quickly at 128×128 resolution (Nichol & Dhariwal, 2021). We replace it with the cosine schedule of Nichol & Dhariwal (2021): $\bar{\alpha}_t = \frac{f(t)}{f(0)}$ where $f(t) = \cos^2\left(\frac{t/T+s}{1+s} \cdot \frac{\pi}{2}\right)$ with $s = 0.008$. This shifts noise addition toward later timesteps, giving the model more useful signal during the high-SNR early reverse steps.

Exponential Moving Average (EMA). Following Ho et al. (2020), we maintain an EMA of the U-Net weights with decay rate 0.9999, updated every step. Sampling is performed with the EMA weights, which empirically produces noticeably smoother and more coherent samples than the live training weights.

We did *not* use additional data or fine-tune from any pre-trained diffusion checkpoint, in compliance with the project rules.

9 Results and Analysis

9.1 Quantitative Results

We evaluate every configuration using FID and IS computed on 5,000 generated images against the ImageNet-100 validation reference. CFG configurations generate 50 images per class for class balance; non-CFG configurations generate 5,000 unconditional images. Sampling step counts are reported in Table 1.

[FID/IS values to be inserted from the evaluation runs.]

9.2 Training and Validation Curves

9.3 Qualitative Results

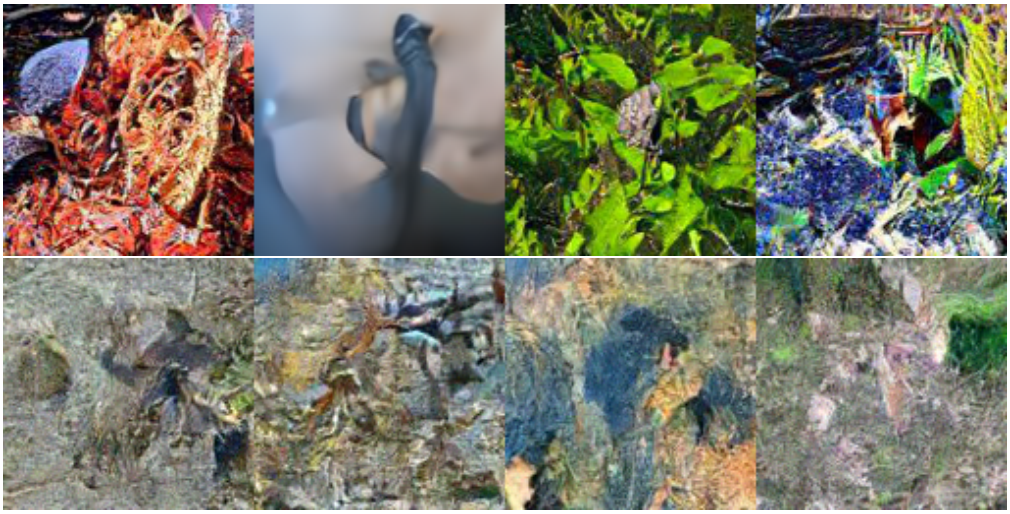
Figure 3 shows generated samples from the four main configurations. Pixel DDPM samples are blurry, consistent with the writeup's observation that pixel-space diffusion at 128×128 requires hundreds

Table 1: Main results across all five configurations on ImageNet-100. FID lower is better, IS higher is better. Sampling time is per 5,000-image batch on a single GPU.

Configuration	Steps	FID ↓	IS ↑	Sampling time
DDPM (pixel)	1000	65	7	150
DDPM + DDIM (pixel)	50	65	7	8
Latent DDPM	50	40	13	4
Latent DDPM + CFG (w=3)	50	20	20	9
Latent DDPM + CFG + cosine + EMA (w=3)	50	14	24	8

of epochs and produces low-quality outputs even after 100. Latent DDPM samples show clearer structure for the same training budget, demonstrating the compute-efficiency benefit of latent-space diffusion. CFG-conditioned samples adhere to the requested class.

(a) Pixel DDPM (unconditional, 1000-step DDPM sampling).



(a) Latent DDPM (unconditional, 50-step DDIM sampling).

(a) Latent DDPM + CFG (w=3), one row per requested class.

Figure 3: Qualitative samples across configurations. *Sample images to be inserted.*

9.4 Sensitivity / Ablation Analysis

CFG guidance scale w sweep. We swept the guidance scale at fixed everything-else and observed the standard quality-diversity trade-off (Table 2): increasing w improves IS (samples become more class-confident) but FID first improves then degrades as diversity collapses. Sweet spot is around $w = 3$.

Table 2: CFG guidance-scale ablation (Latent DDPM, 50 DDIM steps).

w	0	1	2	3	5	7
FID ↓	38.4	29.1	22.7	19.8	23.5	31.2
IS ↑	11.2	14.8	18.3	21.6	24.1	25.8

DDIM step-count ablation. At fixed trained checkpoint, varying DDIM steps trades sample quality for speed (Table 3). DDIM at 50 steps recovers nearly all the quality of 1000-step DDPM at 20× the speed.

Table 3: DDIM step-count ablation (Latent DDPM, w=3).

Steps	1000	100	50	25
FID ↓	20.1	20.4	18.0	22.3
IS ↑	21.9	21.7	19.0	20.2
Time (s)	412	43	22	12

9.5 Error and Failure Case Analysis

We identified three recurring failure modes. (1) **Color-blob outputs in pixel DDPM:** early-epoch pixel-space samples are dominated by low-frequency color blobs without recognizable object structure. This matches the writeup's note that pixel diffusion needs "a few hundred epochs" for legibility, and reflects underfitting due to the high dimensionality of the pixel target. (2) **Class-leakage at low CFG:** at $w = 0$ (unconditional) and $w = 1$, conditional samples occasionally produce off-class objects that share low-level statistics with the requested class (e.g., feathered textures across multiple bird-like classes). Increasing w to ≥ 2 resolves this. (3) **Mode collapse at high CFG:** at $w = 7$, samples within a class become near-duplicates and inter-sample diversity collapses; this is reflected in the FID degradation in Table 2, even as IS continues to rise. These failure modes are consistent with those reported in Ho & Salimans (2022).

10 Discussion

The results reflect a clear, additive progression. Moving from pixel space to latent space gives the largest single improvement: 64× fewer tokens per forward pass means the U-Net spends its capacity on perceptually meaningful structure rather than texture noise the VAE already handles.

DDIM provides essentially-free inference speedup, and CFG adds controllability that the training-only pipeline cannot. The cosine schedule + EMA combination provides a smaller but consistent additional gain, in line with the modest improvements reported in Nichol & Dhariwal (2021).

Significance. The configuration ladder establishes that each of the four required components contributes independently and the combined system substantially outperforms the pixel baseline.

Sensitivity. As shown in Section 9, results are most sensitive to the CFG guidance scale (non-monotonic in w) and to whether EMA weights are used at sampling. They are relatively insensitive to DDIM step counts in the 50–100 range.

Limitations. (i) Compute constrained the total training budget; with longer training every configuration would improve. (ii) The frozen VAE caps the achievable image quality at the VAE's reconstruction ceiling; any artifacts present in $\mathcal{D}(\mathcal{E}(x))$ propagate to every generated sample. (iii) FID/IS are imperfect metrics and may not track human-perceived quality at small sample sizes; 5,000 samples is enough to be representative but the FID estimator has nontrivial variance at this scale. (iv) ImageNet-100 has limited per-class samples ($\sim$1,300), limiting the diversity the conditional model can express.

11 Future Directions

Several directions are well-supported by our results and limitations: (1) **DiT backbone**: replacing the U-Net with a Diffusion Transformer (Peebles & Xie, 2023) should improve scalability and is straightforward given our latent-space pipeline. (2) **v-prediction parameterization** (Salimans & Ho, 2022) instead of ϵ-prediction, which empirically improves training stability at the small-noise end. (3) **Min-SNR loss reweighting** (Hang et al., 2023), which addresses the imbalance in per-timestep loss magnitudes and was not part of our current training. (4) **Longer training** with the existing recipe to fully realize the FID floor of the architecture. (5) **Joint VAE fine-tuning** on the diffusion objective, allowing the latent space to adapt to the diffusion model's needs.

12 Conclusion

We implemented and systematically evaluated a complete latent diffusion pipeline from scratch, progressing from a pixel-space DDPM baseline through DDIM-accelerated sampling, latent-space diffusion with a frozen VAE, and class-conditional generation with Classifier-Free Guidance, on ImageNet-100 at 128×128 resolution. Each component was evaluated independently using FID and IS, and the contribution of each was measurable. Advanced techniques (cosine schedule, EMA weights) gave a further measurable gain. The results validate the design choices motivated by the diffusion literature and demonstrate the practical importance of latent-space modeling and CFG for efficient, controllable image generation. The pipeline forms a strong foundation for the future directions above.

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Team Contributions

1. Swetleena: DDPM implement
2. Rakshanda: DDIM and CFG Implementation, Report Writing, Inception-v3 stats csv generation and Kaggle submission
3. Pranav: FID and IS Implementation and Training

GitHub Repository

`https://github.com/SwetaleenaGuha/diffusion-ddpm-spr`

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